Started on	Sunday, 12 February 2023, 8:45 PM
State	Finished
Completed on	Sunday, 12 February 2023, 8:45 PM
Time taken	10 secs
Marks	0/40
Grade	**0** out of 100

Question **1**

Not answered

Marked out of 1

Which of the following is true concerning multiple regression analysis?

Select one:

- It is a univariate test
- Many dependent variables can be included
- It is a nonparametric test
- It is used to compare proportions
- Many independent variables can be included

Your answer is incorrect.

Multiple regression is used when we have one dependent (Y) variable and many independent (X) variables. The purpose of multiple regression is to find an equation that best predicts the Y variable as a linear function of the X variables. It is a multivariate test that requires parametric assumptions to be satisfied.

The correct answer is: Many independent variables can be included

Question 2

Not answered

Marked out of 1

Which of the following statement is true concerning correlation and regression?

Select one:

- Multiple regression is poßible only if correlation is negative.
- Regression can be positive negative or absent
- Spearman correlation is used for parametric data
- Correlation is a prerequisite for regression
- Correlation is used to predict one value from the other

Your answer is incorrect.

Correlation is a prerequisite for regression. If A and B are not correlated, then we cannot predict A from B or vice versa. Regression is used to predict A from B. Correlation can be positive or negative to produce regression. The values range from -1 to +1. Pearson correlation is used for parametric data while Spearman's is used for non-parametric data.

The correct answer is: Correlation is a prerequisite for regression

Question 3

Not answered

Marked out of 1

Which of the following statistical methods do not adjust for potential confounding variables?

Select one:

- Mantel-Haenszel test
- Least squares method
- Stratification
- Multiple regression
- Restriction

Your answer is incorrect.

The least squares method is a regression method that is not usually employed to tackle confounders.

The correct answer is: Least squares method

Question 4

Not answered

Marked out of 1

Which of the following statistical test is useful if one tries to find out whether the mean IQ of candidates taking MRCPsych exams differ from the mean IQ of candidates taking MRCGP exams?

Select one:

- Chi square test
- Kruskal Wallis test
- Unpaired t test
- Z test
- Paired t test

Your answer is incorrect.

In this example, mean values of two groups are tested. The unpaired t-test is appropriate as the two groups yield independent observations If instead, one tries to compare the proportion of geniuses in the two groups (category - frequency counts) then, chi-square test may be useful.

The correct answer is: Unpaired t test

Question 5

Not answered

Marked out of 1

Which of the following statistical tests would you use when comparing two groups to compare the risk of committing suicide after engaging in two different methods of self-harm?

Select one:

- Kappa test
- Kruskal Wallis test
- Student's t test
- Cohen's d test
- Chi square test

Your answer is incorrect.

The Chi-Square statistic compares the tallies or counts of categorical responses between two (or more) independent groups. (note: Chi-square tests can only be used with counted numbers and not with percentages, proportions, means, etc.)

The correct answer is: Chi square test

Question 6

Not answered

Marked out of 1

Which of the following tests does not require a normal distribution of residuals of the dependent variable?

Select one:

- ANCOVA
- Logistic regression
- ANOVA
- Paired t test
- Unpaired t test

Your answer is incorrect.

Logistic regression can be carried out on ordinal variables, not requiring normal distribution.

The correct answer is: Logistic regression

Question 7

Not answered

Marked out of 1

Which of the following would be a perfect kappa value while testing agreement between two observers for a categorical measure?

Select one:

- 1
- 100
- Zero
- Infinity
- -1

Your answer is incorrect.

A value of kappa equal to +1 implies a perfect agreement between the two raters while that of -1 implies perfect disagreement.

The correct answer is: 1

Question 8

Not answered

Marked out of 1

Which one of the following is a measure of internal consistency of a test?

Select one:

- Inter-rater reliability
- Split half reliability
- Test-retest reliability
- Face validity
- Content validity

Your answer is incorrect.

Split-half reliability is a measure of internal consistency reliability. The items on the scale are divided into two halves, and the resulting half scores are correlated in reliability analysis. High correlations between the halves indicate consistency in reliability analysis.

The correct answer is: Split half reliability

Question 9

Not answered

Marked out of 1

Which one of the following measures is used in correlation analysis for nonparametric data?

Select one:

- Pearson's correlation
- Cronbach's alpha
- Kappa statistics
- Spearman's correlation
- Cohen's d

Your answer is incorrect.

Spearman's: non-parametric, Pearson's: parametric. Kappa is a measure of agreement, not correlation

The correct answer is: Spearman's correlation

Question 10

Not answered

Marked out of 1

A 2X2 contingency table is constructed to analyse the primary outcome data of a trial. The degrees of freedom to use chi-square statistics is

Select one:

- 2
- 3
- 4
- 1
- -4

Your answer is incorrect.

for chi square, df = (number of rows-1) X (number of columns-1)

The correct answer is: 1

Question **11**

Not answered

Marked out of 1

In a cohort of 1064 subjects with an episode of self-harm followed up for the next three years, 20 commit suicide, while 45 die of natural causes. In 11 patients, the verdict is open as the cause of death is unclear. Which of the following statistical test can be used to determine if the suicide risk is significantly higher than expected?

Select one:

- Mann Whitney U Test
- Wilcoxon test
- Cox Regression analysis
- Chi-square test
- One sample T test

Your answer is incorrect.

To compare observed vs. expected proportions (in cells), the chi-square test is the best option.

The correct answer is: Chi-square test

Question **12**

Not answered

Marked out of 1

In statistical terms, the term multivariate often refers to

Select one:

- One or more dependent variables
- Multiple names used for an outcome variable
- Multiple independent variables
- More than two groups being compared
- Same variable measured more than once

Your answer is incorrect.

Multivariate tests often deal with multiple independent variables.

The correct answer is: Multiple independent variables

Question **13**

Not answered

Marked out of 1

Which of the following is an example of nominal data?

Select one:

- Weight expressed in units of kilogram
- Intelligent Quotient
- Score from Beck's Depression Inventory
- Gender
- Temperature expressed using celsius scale

Your answer is incorrect.

Nominal data has no order and allow for only qualitative classification. We can identify a group, but we cannot quantify or rank order the categories. Gender is an example of nominal data.

The correct answer is: Gender

Question **14**

Not answered

Marked out of 1

A novel drug X is compared with an established drug Y for treating a condition that requires long-term hospitalisation. The primary outcome of interest in this study is the time was taken to discharge a patient from the hospital. The researchers are aware that many other factors could affect this outcome in addition to the interventions that are evaluated. Which of the following is the best statistical test that can be used?

Select one:

- ANOVA test
- Cox's proportional hazard test
- Log rank test
- Chi-square test
- McNemar test

Your answer is incorrect.

A Cox proportional hazard test is a statistical technique for exploring the relationship between the survival (time to event function) and several explanatory variables (including the predictors and covariates). This can help to test and find out the variable that has the most important impact on time to the discharge of a patient.

The correct answer is: Cox's proportional hazard test

Question **15**

Not answered

Marked out of 1

You are working as a specialist trainee in a liaison psychiatry unit. Your consultant supervisor has recently assessed a patient with pancreatic cancer for symptoms of depression. He describes the patient's depression as "severe". What kind of data does this description provide?

Select one:

- Qualitative
- Ordinal
- Binary
- Ratio
- Interval

Your answer is incorrect.

Clinical grading of the degree of depression is ordinal, just like TNM staging for cancer.

The correct answer is: Ordinal

Question 16

Not answered

Marked out of 1

Which of the following statistical test allows the prediction of the magnitude of one (dependent) variable from another (independent) variable?

Select one:

- Analysis of variance
- Efficacy analysis
- Intention to treat analysis
- Correlation analysis
- Regression analysis

Your answer is incorrect.

Using regression, we can assess whether the value of an independent variable can be used to predict the value of a dependent variable. For example, using weight (the independent variable) to predict blood cholesterol levels (the dependent variable). In simple linear regression, a "best fit" straight line is determined for the data in which the dependent variable is plotted on the y-axis versus the independent variable on the x-axis.

The correct answer is: Regression analysis

Question 17

Not answered

Marked out of 1

Which of the following is a necessary but not sufficient condition to construct a regression analysis?

Select one:

- Existence of temporal relationship between the variables
- P value of correlation coefficient is <0.05
- Existence of causal relationship between variables
- Existence of positive r value in correlation analysis
- Existence of significant correlation between variables

Your answer is incorrect.

The dependent and independent variable must have a correlation to construct a regression equation. This does not mean they must be causally related or temporally related. The correlation can be in positive or negative values.

The correct answer is: Existence of significant correlation between variables

Question **18**

Not answered

Marked out of 1

Interval data can be analysed using all of the following statistical tests except

Select one:

- Multiple regression
- T test
- ANOVA
- Linear regression
- Chi-square

Your answer is incorrect.

Interval data cannot be analyzed with Chi-square unless they are converted into categories initially. All other tests can be used when parametric conditions are satisfied.

The correct answer is: Chi-square

Question **19**

Not answered

Marked out of 1

A drug developer is interested in the tendency of SSRIs to cause GI bleeding. To examine the potential of escitalopram in causing intestinal haemorrhages, she develops an injectable form of escitalopram. 30 rats receive daily injections for three weeks while another group of 25 rats are given saline injections for three weeks. The intestines are examined after sacrificing the rats 3 hours after last injection on the 3rd week. It is noted that out of 30 escitalopram-rats, 12 had mucosal haemorrhages. In contrast, only three saline-rats showed mucosal haemorrhages. Which of the following tests would be the best to examine the relationship between escitalopram and mucosal haemorrhage?

Select one:

- Pearson correlation coefficient
- Linear regression
- Analysis of variance (ANOVA)
- Paired t test
- Chi-square test

Your answer is incorrect.

To study the relationship between categories in a contingency table, one can construct a ratio of observed vs. expected frequencies in the cells. This is called chi-square test. It is a nonparametric method of testing the significance of proportions across groups.

The correct answer is: Chi-square test

Question 20

Not answered

Marked out of 1

A recently reported trial in BMJ investigated the effects of three variables (chronicity, duration of untreated psychosis and cannabis exposure) on the efficacy of a newly developed antipsychotic drug. It reported a significant ($p < 0.001$) analysis of variance (ANOVA) result with a large two-way interaction. Which of the following is the best interpretation of these results?

Select one:

- The drug has no efficacy as there is a two way interaction
- All three variables are necessary to explain the drug's efficacy
- Two of the three variables affect the efficacy
- Only one of the three variables affects the drug's efficacy
- The three variables interact among themselves but have no effect on the drug's efficacy

Your answer is incorrect.

The term "two-way interaction" indicates that two of the three variables in the study affect the efficacy of the drug. For example, persons with less chronicity and lower duration of treated psychosis (DUP) may benefit the most from the drug, regardless of cannabis effect. A one-way interaction indicates that any of the three variables in isolation can affect the drug's efficacy.

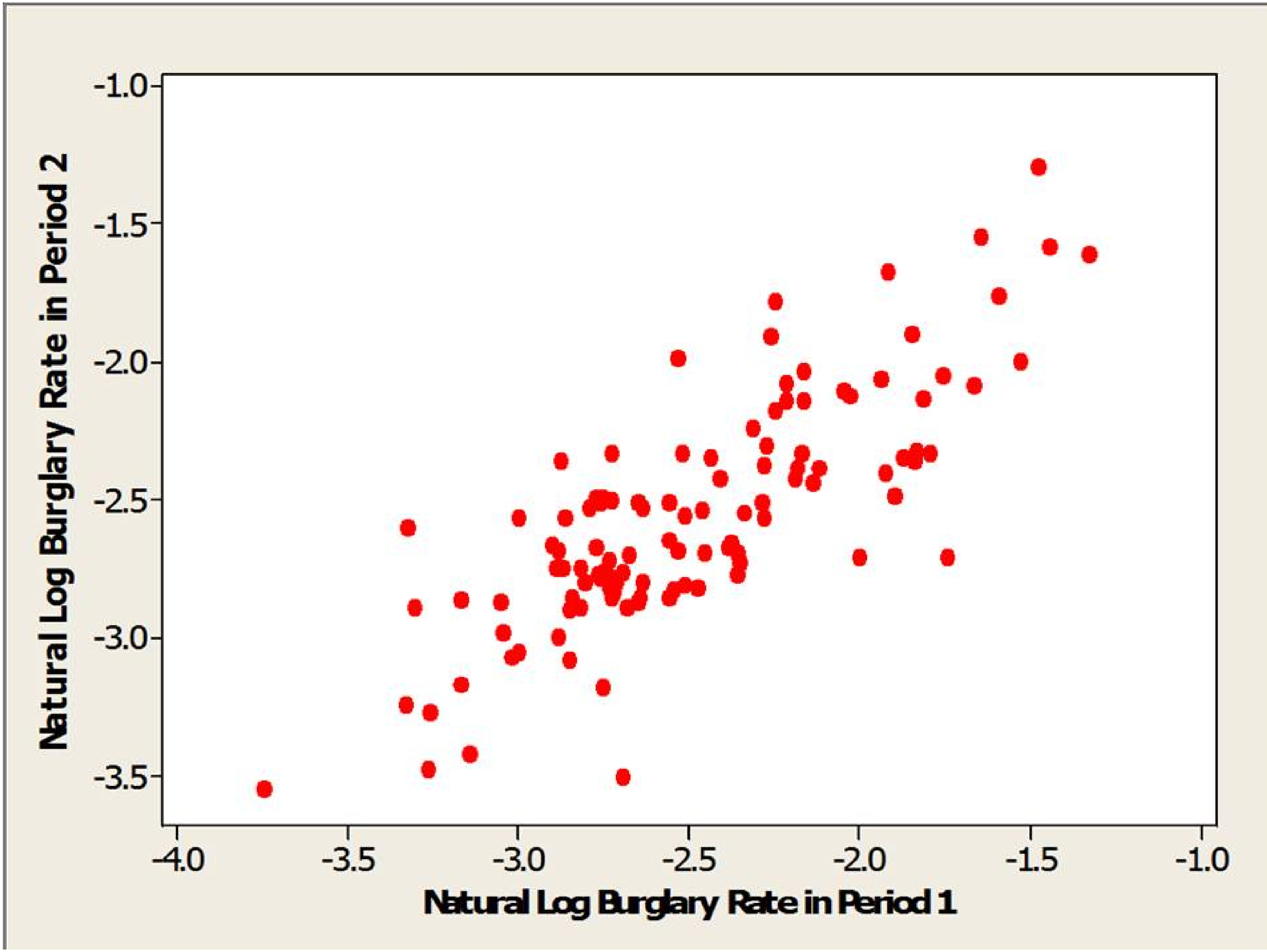
The correct answer is: Two of the three variables affect the efficacy

Question 21

Not answered

Marked out of 1

The attached graph was presented in a study comparing the probability of burglary in various households in a city before and after an intervention. Which of the following statistical methods has been used?


-1.0
-1.5
-2.0
-2.5
-3.0
-3.5
Natural Log Burglary Rate in Period 2
-4.0
-3.5
-3.0
-2.5
-2.0
-1.5
-1.0
Natural Log Burglary Rate in Period 1

Select one:

- Spearman's correlation
- Multiple regression
- Logistic regression
- Meta-analysis
- Simple linear regression

Your answer is incorrect.

Logistic regression is a statistical method analogous to linear regression for the analysis of categorical data. In logistic regression, the dependent variable is categorical (in the simplest case, dichotomous, or having two outcomes like absence or presence of disease). The independent variables can be categorical, or else continuous or discrete. Another very useful feature of logistic regression is the ability to calculate odds ratios of an outcome of the dependent variable (e.g., depression) given specified values of one of the independent variables (e.g., alcohol use vs. no use) while controlling for the values of the other independent variables (e.g., smoking, family history, and work stress). Tilley N., Pease K., Hough M. and Brown R. (1999) Burglary Prevention: Early Lessons from the Crime Reduction Programme, Crime Reduction Research Series Paper 1: Home Office, London.

The correct answer is: Logistic regression

Question **22**

Not answered

Marked out of 1

Which of the following tests could be used to study statistical association from the above data

Select one:

- Chi square test
- Kruskal Wallis test
- Pearson's correlation test
- One sample t test
- Student's t test

Your answer is incorrect.

A Chi-square test for covariates in a logistic regression equation can be used to test the null hypothesis that all the regression coefficients in the model are zero. A significant result suggests that at least one covariate is significantly associated with the dependent variable.

The correct answer is: Chi square test

Question 23

Not answered

Marked out of 1

Which of the following data is best analysed using Wilcoxon rank sum test?

Select one:

- Continuous data from two independent non-normal samples
- Means from two independent normally distributed samples
- Ordinal data from two paired non-normal samples
- Ordinal data from two independent normally distributed samples
- Means from two paired observations

Your answer is incorrect.

We can use three different tests for paired data, the Wilcoxon rank sum, sign and paired t tests. If the differences are Normally distributed, the t-test is the most powerful test. The Wilcoxon test is useful for non-Normal data from large samples. The sign test is similar in power to the Wilcoxon for very small samples, but as the sample size increases the Wilcoxon test becomes much more powerful. In the Wilcoxon test, the difference between each pair of data is determined and then ranked. One assigns a positive or negative sign to the ranking, depending on the sign of the original difference. The positive and negative ranks are summed separately and are used to determine the level of statistic significance using a special table.

The correct answer is: Ordinal data from two paired non-normal samples

Question **24**

Not answered

Marked out of 1

A study was designed to determine the ability of providers (medical residents and nurse practitioners) in inpatient cardiac units to diagnose clinically significant depression among a cohort of 25 patients admitted with myocardial infarction. Patients underwent standard diagnosis with the Standardized Clinical Instrument for DSM (SCID) applied by a research psychiatrist. The providers identified 1 of 9 subjects (11.1%) with current major depression. To examine significant differences in proportions of diagnosed depression between providers and research psychiatrist a chi-square distribution was used. How many degrees of freedom should be employed for the calculation?

Select one:

- 3
- 0.11
- 2.32
- 1
- 4

Your answer is incorrect.

Degrees of freedom indicate the extent to which a set of observations are 'free' to vary; they can be calculated as the sample size minus the number parameters (constraints) that have to be estimated. If you are asked to choose a set of three numbers which add up to make a particular total (say, 100). You can choose two random numbers less than 100, but the third choice is 'constrained' by the previously chosen two numbers. In other words, two of the numbers are 'free' to take any value, but the remaining number is fixed by the constraint imposed by the total value. Therefore, the numbers have two degrees of freedom when calculating a sum. For chi-square test, this can be calculated by the formula df = (number of columns-1) X (number of rows - 1). Here this is df = 1X1 = 1. For a 3 × 3 table there are (3 - 1)(3 - 1) = 4 degrees of freedom.

The correct answer is: 1

Question **25**

Not answered

Marked out of 1

Which of the following test is used to compare statistically the survival of two groups for a single variable of interest?

Select one:

- Mann Whitney U test
- ANOVA
- Log rank test
- Logistic regression
- T test for unpaired observation

Your answer is incorrect.

Comparison of two survival curves can be done using a statistical hypothesis test called the log-rank test. It is used to test the null hypothesis that there is no difference between the population survival curves (i.e. the probability of an event occurring at any time point is the same for each population). Retrieved from Critical Care 2004, 8:389-394

The correct answer is: Log rank test

Question **26**

Not answered

Marked out of 1

An alcohol rehabilitation unit is running a self-enhancing group for alcohol users. To measure the efficacy of their intervention, a self-esteem scale is administered to all patients at the baseline before entering the 12-week treatment. The same scale is reapplied on completion. Choose the most appropriates statistical test:

Select one:

- Paired t test
- Chi square test
- MannWhitney U test
- Unpaired t test
- McNemar's test

Your answer is incorrect.

Most psychiatric questionnaires can be treated as continuous data with assumed parametric distribution. As this study involves two tests/ subject - there are paired observations. As means are compared, a paired t-test would be most appropriate.

The correct answer is: Paired t test

Question 27

Not answered

Marked out of 1

Patients on discharge from male and female wards were given a patient satisfaction questionnaire that rated satisfaction from a hospital stay in a scale of 0 to 9. An NHS manager is interested in comparing the difference in satisfaction between male and female ward inpatients. Which of the following test is most suitable?

Select one:

- Unpaired t test
- McNemar's test
- Mann Whitney U test
- Chi square test
- Wilcoxon matched pair test

Your answer is incorrect.

This is ordinal data - and it is unpaired. The Mann-Whitney U test is a nonparametric test that is used to determine whether two sets of data are significantly "different." As noted in the question, the variable being measured is ordinal. The two sets of data are assumed to be independent and randomly drawn (males vs. females). The statistics of interest in this test are the medians of both sets of data, and the test determines whether or not the difference in the medians is statistically significant at a given level.

The correct answer is: Mann Whitney U test

Question 28

Not answered

Marked out of 1

A study was carried out to measure attitude about mental health problems in 60 relatives of mentally ill patients. 56 controls matched for age and sex were also selected. The results were categorised into positive and negative attitudes. Which of the following tests is suitable to analyse the above data?

Select one:

- Wilcoxon test
- ANOVA (one way)
- Chi square test
- McNemar's test
- Unpaired t test

Your answer is incorrect.

The chi-square statistic is a nonparametric test of bivariate count or frequency data. This examines the disparity between the actual count frequencies and the expected count frequencies when the null hypothesis of no difference between the groups is true. In this question, we can apply the chi-square test to determine whether an association exists between the family history of mental illness and positive attitude.

The correct answer is: Chi square test

Question 29

Not answered

Marked out of 1

A researcher is interested in studying the effect of an MCQ book on the test performance of trainee psychiatrists. He designs a controlled trial in which subjects in group A attempt a pre-test, read the book and attempt a post-test. Subjects in group B attempt a pre-test, 'prepare as usual' without reading the new book and attempt a post-test. Which one of the following is the most appropriate method of analysing the outcome?

Select one:

- Unpaired t test of post-test scores in group A and B
- Paired t test comparing mean score of all subjects in pre-test and post-test
- Unpaired t test of mean differences in pre and post-test scores between two groups
- Correlation analysis of pre-test and post-test scores
- Paired t test of mean differences in pre and post-test scores between two groups

Your answer is incorrect.

In this question, there are two groups - A and B. The researcher would probably compare the mean 'gain' of scores between pre- and post-test conditions in two groups. Though paired observations are made, these observations are not compared directly. Instead, differences in paired means are compared. This is similar to testing two group means - hence an unpaired t-test would be apt.

The correct answer is: Unpaired t test of mean differences in pre and post-test scores between two groups

Question 30

Not answered

Marked out of 1

The use of Chi-square test is not valid under which of the following conditions?

Select one:

- The observed cell frequency is more than 5 in 20% or more of the cells
- The observed cell frequency is less than 5 in 20% or more of the cells
- The observed cell frequency is less than 20 in 5% or more of the cells
- The expected cell frequency is less than 5 in 20% or more of the cells
- The expected cell frequency is more than 20 in 5% or more of the cells

Your answer is incorrect.

To apply chi-square, each individual must be represented only once in the study (independent samples). The rows (and columns) of the table must be mutually exclusive. Also, Cochran's criteria should be fulfilled if the chi-square test statistic is to be used for testing statistical significance. These criteria are (1) All expected values in each cell have a frequency count =1. (i.e. non-zero values) (2) at least 80% of total cells must have an expected value of =5.

The correct answer is: The expected cell frequency is less than 5 in 20% or more of the cells

Question 31

Not answered

Marked out of 1

Which of the following data is better suited for weighted kappa compared to Cohen's Kappa for measuring agreement?

Select one:

- Categorical, non-binary data
- Parametric data
- Binary data
- Ordinal data
- Continuous data

Your answer is incorrect.

For ordinal data, we can calculate a weighted kappa which takes into account the extent to which the observers disagree as well as the frequencies of the agreement. The weighted kappa is very similar to the intraclass correlation coefficient

The correct answer is: Ordinal data

Question **32**

Not answered

Marked out of 1

Which of the following statistical tests does not assume normality of data distribution

Select one:

- Paired t test
- Analysis of variance
- Multiple linear regression by least squares
- Pearson's coefficient
- Mann Whitney U Test

Your answer is incorrect.

Tests that make no assumption of normality are termed nonparametric. Examples of nonparametric tests include the Wilcoxon signed rank test, the Mann-Whitney U test, the Kruskal-Wallis test, and the McNemar test.

The correct answer is: Mann Whitney U Test

Question 33

Not answered

Marked out of 1

Which of the following is false with regard to parametric statistics?

Select one:

- These are best used for small sample sizes
- The data must be normally distributed
- The observations must be independent
- Skewed data can be transformed to be adapted for parametric statistics
- These methods treat observations as if they come from same samples.

Your answer is incorrect.

For measured quantitative variables (continuous, ratio or interval data), it is generally assumed that the data follow a bell-shaped curve and that the statistics focus on the center and width of the curve. These are the so-called parametric statistics. These methods treat observations as if they come from the same homogenous sample. Note that under certain circumstances even skewed (not normal) data can be analyzed using parametric statistics following transformation methods (see below). The conditions for using parametric statistics include 1. The observations must be independent 2. The observations must be drawn from normally distributed populations 3. These populations must have the same variances

The correct answer is: These are best used for small sample sizes

Question 34

Not answered

Marked out of 1

Which of the following is NOT a necessary assumption when using Analysis of Variance technique?

Select one:

- The populations are normally distributed
- Parametric assumptions are met
- The mean is equal to zero
- The samples are independently selected
- The variances of the populations are the same

Your answer is incorrect.

Assumptions for using ANOVA include 1. Parametric distribution (i.e., normal distribution of observations in groups) 3.Equal variance among the tested groups 3.Independent observations

The correct answer is: The mean is equal to zero

Question **35**

Not answered

Marked out of 1

Which of the following tests is a non-parametric equivalent of comparing two independent means using t-tests?

Select one:

- Kruskal Wallis test
- Wilcoxon test
- Chi square test
- Spearman's test
- Mann-Whitney U test

Your answer is incorrect.

Wilcoxon rank sum test is used for comparing two paired observations in nonparametric fashion; thus it is equivalent to paired t. Mann-Whitney U test is used for independent observations in two groups. Kruskal-Wallis is more or less an ANOVA equivalent used for 3 or more groups. Spearman's correlation is the non-parametric equivalent of Pearson's.

The correct answer is: Mann-Whitney U test

Question **36**

Not answered

Marked out of 1

Which of the following methods can be used to correct for multiple testing in a dataset?

Select one:

- Bonferroni correction
- Increasing p value
- Non-parametric testing
- Scatter plotting
- Cohen's kappa

Your answer is incorrect.

Bonferroni correction: To correct for this multiple testing Bonferroni method can be used - it is rather rigorous and strict and may lead to many false negatives. According to this the significance level for multiple-tested data is altered as (normal significance level/number of statistical analyses carried out). Hence if p = 0.05 normally, and if ten subanalyses are attempted, then p becomes 0.005. This may be an overcorrection especially when outcomes are associated with one another. Bonferroni treats each outcome as an individual event.

The correct answer is: Bonferroni correction

Question **37**

Not answered

Marked out of 1

A cohort study reports the number of myocardial infarctions developing in patients on olanzapine. If the study runs for a five-year period, there may still be few myocardial infarctions that may occur after five years. Such unobserved data in survival analysis is called

Select one:

- Hazard data
- Censored data
- Drop out
- Violated data
- Attrition

Your answer is incorrect.

In longitudinal cohort studies, often the outcome of interest is how long it takes for an event to occur e.g. time for relapse, etc. Often the length of time survived by 50% of a sample without achieving an outcome (usually adverse) is used - called median survival rate. This data can be plotted as life tables or Kaplan-Meier survival curves. Here the time is plotted along the x-axis, and proportion surviving is plotted along the y-axis. These curves have a stepped appearance, with each step taking place at a time of observation. If the study runs for a 5-year period, then not all subjects would have reached the outcome by that time. Hence, these observations are censored.

The correct answer is: Censored data

Question 38

Not answered

Marked out of 1

You are comparing means of two sets of scores that are repeated observations on the same group of individuals. The scores are normally distributed. Which test will you use?

Select one:

- Paired t test
- Unpaired t test
- ANOVA
- Wilcoxon rank sum test
- MANOVA

Your answer is incorrect.

The unpaired t-test compares independent samples where observation is made only once from each in a sample. In a paired t-test, sample means are compared to the same group i.e. pretest vs. posttest. Here each subject is observed twice, say, before and after an intervention and the scores are compared for statistical significance.

The correct answer is: Paired t test

Question 39

Not answered

Marked out of 1

An investigator is interested in studying the relationship between the number of long-term relationships one had by the age of 40 and perceived quality of life using QOL scores. 178 households were systematically sampled, and 125 individuals over the age 40 were given QOL questionnaire. It was found that the number of long-term relationships by the age of 40 ranged from 0 to 6. Choose two tests

Select one:

- Spearman correlation and factorial ANOVA
- Pearson's correlation and one way ANOVA
- Spearman correlation and repeated measure ANOVA
- Pearson's correlation and MANOVA
- Spearman correlation and one way ANOVA

Your answer is incorrect.

Spearman's rank correlation is a nonparametric analog to Pearson's correlation coefficient. It is used to test if there is an association between two mutually dependent variables which can be continuous or discrete. Here 'number of relationships by the age of 40' is a discrete data. ANOVA is used with data when the dependent variable is continuous and all independent variables are categoric (we can separate the data in question into 4 categories those with no relationships, 1 - 2 relationships, 3 - 5 and >5 etc.). One-way ANOVA is used when there is one categoric independent variable, whereas N-way ANOVA is used when there is more than one.

The correct answer is: Spearman correlation and one way ANOVA

Question **40**

Not answered

Marked out of 1

A psychologist studied the effects of birth order (first/only, middle, youngest) of men vs. women in their degree of neuroticism. The measure of neuroticism averaged responses across 15 items that asked for 5-point Likert ratings. Which of the following statistics is most appropriate for this analysis?

Select one:

- One-way ANOVA.
- Unpaired t test
- Kruskal-Wallis test.
- Friedman ANOVA for ranks.
- Factorial ANOVA.

Your answer is incorrect.

Multiple variables here - 3 birth orders X 2 genders. A factorial ANOVA would be more appropriate than other listed.

The correct answer is: Factorial ANOVA.